

PROJECT DESIGN PHASE

Problem – Solution Fit Template

Date	15 February 2026
Team ID	T-12
Project Name	FinVault – Virtual Stock Trading Platform
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.

1. CUSTOMER SEGMENT(S) CS - High school and college students learning economics. - Retail trading beginners seeking risk-free practice. - Financial literacy educators hosting mock trading labs.	6. CUSTOMER CONSTRAINTS CC - Zero or limited real-world capital. - Age restrictions (under-18 cannot open brokerage accounts). - Short attention span requiring simple dashboards.	5. AVAILABLE SOLUTIONS AS - Live brokerage accounts (high risk of capital loss). - Excel spreadsheets or paper trading logs (tedious). - Institutional terminal simulators (complex, steep learning curve).
2. JOBS-TO-BE-DONE / PROBLEMS J&P - Practice placing stock market orders (BUY/SELL). - Analyze historical stock prices and visual trends. - Real brokerages are intimidating and require capital.	9. PROBLEM ROOT CAUSE RC Financial markets have a high barrier to entry due to the threat of capital loss and strict regulatory requirements for open trading accounts. There is a lack of accessible, safe, and intuitive software tailored for educational learning, forcing beginners to learn by risking real money or using dry textbooks.	7. BEHAVIOUR BE Beginners watch trading tutorials on YouTube, join online investing forums, discuss stock tips with peers, or attempt to track mock portfolios manually on spreadsheets or notes apps.
3. TRIGGERS TR - Financial news reports on stock price movements. - Social media stock trading trends and discussions. - Educational assignments on basic investing concepts.	10. YOUR SOLUTION SL FinVault: A web-based MERN trading sandbox. Provides a virtual wallet (\$100k), instant BUY/SELL execution, interactive Chart.js stock histories, customizable profile uploads (Multer), and admin dashboard controls.	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE: - Financial blogs and YouTube trading video courses. - Online discussion channels (Reddit, Discord groups). - Spreadsheet templates (Excel/Google Sheets). 8.2 OFFLINE: - College lectures and high school business classes. - Peer discussions and finance club meetups. - Hand-written watchlists and trading logs.
4. EMOTIONS: BEFORE / AFTER EM Before: Fearful of financial loss, overwhelmed by complex terminology, confused by trading charts. After: Empowered, confident in placing trades, clear understanding of portfolio metrics.		

Problem-Solution Fit Analysis & Key Takeaways:

FinVault bridges the gap between the user's desire to learn stock trading and the barriers preventing active participation (lack of capital and fear of financial loss). By examining the customer canvas, the platform aligns its technical features directly to the target segment's behavior and constraints.

First, the **wallet initialization of \$100,000 mock cash** bypasses the capital constraint (CC) and removes the fear of loss (EM). This transforms the user's negative emotions before using the app (anxiety, confusion) into positive educational outcomes (confidence, financial literacy). Second, the **interactive price history line graph (Chart.js)** and **dynamically filtered search list** map directly to the user's jobs-to-be-done (J&P) by supplying immediate market visualization, which replaces dry textbook learning with an authentic simulated trading desk.

Additionally, the inclusion of a **fully secured administrator dashboard** provides institutions (e.g. schools and colleges) with the controls required to host organized mock trading simulations, unlocking a B2B white-label revenue stream. By offering stateless JWT authorization and atomic database operations, FinVault proves that a sandbox platform can still deliver industry-standard security and consistency, achieving a comprehensive Problem-Solution fit.

References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>